ABSTRACT

of the PhD thesis by Yesmukhamedov Nurmaganbet Seitkaliuly on «Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy», submitted for the degree of Doctor of Philosophy (PhD) in the educational programme (EP) 8D06102 — Computer and Software Engineering

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General characteristics of the research

This dissertation investigates and formalises the integration of fundus image enhancement (preprocessing) with convolutional-neural-network (CNN) classification for automated multi-stage diabetic retinopathy (DR) diagnosis. Its central idea is that preprocessing is an integral component of the diagnostic model rather than ancillary data preparation, because it defines the feature space available to the network. To operationalise this idea, an 8-stage preprocessing pipeline is specified and embedded in the model, and its effect is placed under a controlled experimental contrast against an equivalent baseline trained without the pipeline. The approach is validated across eight public and clinical fundus datasets acquired with cameras of four manufacturers, covering diagnostic dominance, component ablation, direct measurement of domain-shift reduction, cross-dataset transfer, explainability, external clinical performance, and device domain shift, and is oriented towards automated DR screening in resource-limited settings, including the healthcare infrastructure of Kazakhstan.

Relevance of the research topic

Diabetic retinopathy (DR) is one of the leading causes of preventable blindness among the working-age population worldwide. Because the early stages of DR are clinically asymptomatic, timely detection depends on regular screening of fundus images, yet the number of ophthalmologists is insufficient to manually examine the rapidly growing population of diabetic patients — a shortage that is especially acute in resource-limited and rural healthcare settings, including the regions of Kazakhstan. Automated diagnosis based on convolutional neural networks (CNNs) is therefore a practically significant direction of research.

The dominant tradition in automated DR diagnosis — here designated paradigm **P1**, the end-to-end CNN paradigm — treats image preprocessing as ancillary data preparation that does not require methodological discussion, on the assumption that a sufficiently deep network learns the relevant invariances directly from raw or minimally normalised pixels. In practice, however, fundus images acquired by different cameras, under different illumination conditions, and with different noise levels exhibit substantial domain variability. This variability manifests as distribution shift in the input feature space and degrades the generalisation of CNN models trained on one domain when they are applied to another. The methodological gap motivating this dissertation is the absence of a formalised, experimentally controlled treatment of preprocessing as an integral component of the diagnostic model.

This work advances an alternative paradigm — **P2**, the integrated preprocessing-CNN paradigm — in which preprocessing is an integral component of the model itself, because it defines the feature space available to the network and therefore co-determines what the network can learn. The relevance of the topic follows from the need to reduce inter-device and inter-acquisition domain variability while preserving diagnostically relevant retinal features, and to do so under the computational constraints characteristic of real screening environments.

Aim of the research

To develop and experimentally validate an integrated fundus image enhancement and CNN-based classification framework for automated multi-stage diabetic retinopathy diagnosis, in which an 8-stage preprocessing pipeline is specified as an integral component of the model rather than as preparation of the data, and to establish, under controlled contrast against an equivalent configuration trained without that pipeline, what difference the specification makes to five-class diagnostic performance, to transferability and to the interpretability of the resulting model under constrained computational conditions.

Objectives of the research

1. To analyse the problem domain and formulate the research problem: the clinical grading of the disease, the screening requirements of resource-limited settings, the sources and measured impact of fundus image quality loss and its device-specific component, and the methodological practice of existing automated screening systems (Chapter 1).
2. To specify the integrated preprocessing-CNN methodology: the 8-stage pipeline stage by stage with the theoretical foundations of each (histogram equalisation and the dual-constraint CLAHE clip limit, spatial filtering, illumination correction), the classification architectures (ResNet-50, EfficientNet-B3) and their adaptation to five-class grading, the pretraining and fine-tuning strategy, the explainability formalism (Grad-CAM, ALO, IoU), and the evaluation and statistical protocol (Chapter 2).
3. To evaluate the framework experimentally through eight investigations conducted on a common substrate — integrated-pipeline dominance under a 2×2 factorial design (Experiment 1); cumulative component ablation with the CLAHE and flat-field parameter sweeps (Experiment 2); direct measurement of source-to-target distance in feature space across six external corpora; cross-dataset transferability (Experiment 3); Grad-CAM explainability against expert lesion annotation (Experiment 4); external clinical performance (Experiment 5); device domain shift (Experiment 6); and training in a data-scarce regime (Experiment 7) — together with the statistical validation, the comparative placement of the results, and the limitations and boundary conditions of the approach (Chapter 3).

4. To build and describe a screening system around the model, suited to settings without inference acceleration and with intermittent connectivity, with telemedicine and eHealth integration applicable to Kazakhstan healthcare infrastructure, and to state what of it exists and what remains specification (Chapter 4).

Object of the research

The process of automated multi-stage diagnosis of diabetic retinopathy from colour fundus photographs by means of convolutional neural networks. The process is studied in its entirety, from the image as it leaves the camera to the five-class grade the model assigns, rather than in the classification step alone.

Subject of the research

Methods and models for integrating fundus image preprocessing (enhancement) with CNN classification, and the properties the resulting integrated configuration exhibits: diagnostic performance under five-class grading, transferability to corpora not seen during training, alignment between its attention and expert-annotated pathology, and behaviour across camera domains. It is the integration that is manipulated, not the preprocessing considered apart from the network it feeds.

Methodology and methods of the research

The research methodology combines image-processing theory, deep-learning theory, and an experimentally controlled validation framework. The methods used include: optic-disc and fovea localisation for rotation normalisation by a pre-trained, frozen heatmap-regression detector, not co-trained with the classifier, so that the preprocessing remains a fixed transform; adaptive flat-field illumination correction applied inside the FOV mask only, dual-constraint Contrast-Limited Adaptive Histogram Equalisation (CLAHE) in the LAB colour space, CNN classification with ResNet-50 and EfficientNet-B3 backbones, transfer learning from ImageNet for the baseline arm and in-domain pretraining for the integrated arm — the choice among candidate initialisations being settled by a frozen-backbone linear-probe acceptance gate rather than assumed — Focal Loss for class imbalance, Grad-CAM explainability analysis with Attention–Lesion Overlap (primary) and IoU (secondary), maximum mean discrepancy over penultimate-layer features for the direct measurement of domain shift (with per-channel histogram divergence as informational context), and a statistical validation framework based on 5-fold patient-level stratified cross-validation, McNemar and DeLong tests, mixed-effects modelling, bootstrap confidence intervals, and Bonferroni/Holm correction for multiple comparisons.

Empirical (experimental) basis

Eight datasets organised into functional tiers: EyePACS (~35,126 graded images, primary training and ablation, Canon CR-1), whose second, unlabelled split

(~53,576 images) supplies the in-domain pretraining corpus of the integrated arm, disjoint from it by image and patient identity; APTOS 2019 (~3,662, cross-dataset transfer); IDRiD (516 images, 81 with pixel-level lesion masks, clinical validation and explainability, Kowa); Messidor-2 (~1,748, external clinical evaluation, Topcon); DDR (~13,673, device shift, Canon/Topcon); ODIR-5K (~5,000, device shift, Canon/Zeiss); RFMiD (~3,200, device shift, Topcon/Kowa); and a Kazakh clinical dataset (60 images, 30 patients × 2 eyes, balanced by grade, training in a data-scarce regime), which cannot be redistributed, so that results depending on it are not externally reproducible. The device-domain corpora annotated under multi-disease taxonomies are used as sources of DR labels only. Image quality is evaluated with contrast-to-noise ratio (CNR), image entropy, and SSIM; results are reported per corpus and never aggregated into a single figure of merit.

Scientific novelty

1. An 8-stage integrated fundus preprocessing pipeline is formalised as a binding component of the diagnostic model (operationalisation of paradigm P2), comprising canonical flip, OD-fovea rotation normalisation, FOV crop with isotropic resize and centred zero-padding, FOV mask generation as a 4th input channel, adaptive flat-field correction ($\sigma = 0.07 \times$ FOV diameter), dual-constraint stochastic CLAHE on the LAB L-channel, augmentation, and dataset-specific channel-wise normalisation.
2. The hypothesis of integrated-pipeline dominance is formalised and tested under a controlled 2×2 factorial design (configurations A–D) over two established architectures (ResNet-50, EfficientNet-B3), with a pre-registered dominance criterion (Δ weighted F1 $\geq$ 5 pp, Δ ROC-AUC $\geq$ 0.02, no Cohen's Kappa degradation).
3. A cumulative ablation under a single initialisation separates the preprocessing contribution from the initialisation with which the factorial design necessarily confounds it; the decomposition does not convert that design into a single-factor manipulation, and no claim about the isolated effect of preprocessing rests on it.
4. A dual-constraint stochastic CLAHE variant is formalised and validated within the five-class DR context (LAB L-channel; clip limit CL = min(clip_factor × tile_area / 256, global_threshold × tile_area); 80% train-time probability), the stage serving at once as an enhancement operator and as a source of regularisation. The formulation extends the candidate's previously published work on fundus image enhancement rather than originating here; new are its formalisation, its embedding in a specified pipeline, and its validation in the five-class grading task.
5. An adaptive flat-field illumination correction is introduced whose Gaussian σ scales with the per-image FOV diameter rather than using a fixed global value, applied only inside the FOV mask so that the padding is not corrupted.
6. An explicit binary FOV mask is introduced as a dedicated pipeline stage and appended as the 4th input channel, informing the CNN of valid pixel regions and preventing padding artefacts from being learned as features.

7. Attention–Lesion Overlap (ALO) is introduced as a primary, asymmetric explainability metric that directly measures the fraction of a lesion covered by model attention, complemented by Intersection-over-Union (IoU) as a secondary metric, evaluated against IDRiD pixel-level lesion masks. It establishes alignment between model evidence and expert annotation, not clinical localisation of pathology.
8. The mechanism postulated by the central hypothesis is measured directly rather than inferred from its consequence: source-to-target distance is computed at the penultimate layer under both configurations across six external corpora, at the cost of forward passes only. Two design features carry the contribution — normalisation uses source-domain statistics and is never recomputed on the target, so the measurement is a property of the preprocessing and not of a procedure fitted to the target; and its falsifiability was recorded before it was made, since two stages of the pipeline are bound to the source domain by construction and a reversal was therefore a live possibility.
9. The integrated arm is initialised from in-domain rather than natural-image pretraining, with the choice among candidate initialisations settled by a frozen-backbone linear-probe acceptance gate rather than by assumption. The negative outcome is reported as evidence: label-free self-supervision trained from scratch on the in-domain corpus failed that gate, across several protocols of the same family and without improvement from longer training, and was not admitted.
10. The integrated pipeline is validated across a tiered multi-dataset, multi-device architecture (EyePACS, APTOS 2019, IDRiD, Messidor-2, DDR, ODIR-5K, RFMiD, and a Kazakh clinical dataset), organised by function and spanning four camera manufacturers (Canon, Topcon, Kowa, Zeiss), including cross-dataset transfer, external clinical performance, and device domain shift.
11. A defect shared by three measures in common use for expressing external robustness is identified analytically: the degradation quantity, the generalisation ratio and the retention ratio each normalise or difference an arm's external performance against that same arm's in-domain performance, and therefore penalise a configuration for its in-domain strength. The analysis is strictly descriptive and rehabilitates no result of this work.
12. Two contributions are non-empirical by design: the coupled thermal-optical model of fundus tissue response under laser exposure, a theoretical and computational contribution only; and the modular screening-system architecture, a design contribution realised in part as a working demonstrator, which establishes realisability and operating behaviour and is evidence for no diagnostic claim — no clinical deployment testing conducted.

Main results of the research

1. A critical analysis of the medical, epidemiological, and technical context of automated DR diagnosis and of fundus image quality variability was carried out; the limitations of the end-to-end paradigm (P1) were identified and the research problem was formulated.

2. An 8-stage integrated preprocessing pipeline was designed and formalised as an integral component of the diagnostic model (paradigm P2), comprising canonical flip, OD-fovea rotation normalisation, FOV crop with isotropic resize and centred zero-padding, FOV mask generation as a 4th input channel, adaptive flat-field correction, dual-constraint stochastic CLAHE, augmentation, and dataset-specific normalisation.
3. The dominance of the integrated configuration over the baseline was established under a controlled 2×2 factorial design (configurations A–D) on EyePACS for both ResNet-50 and EfficientNet-B3: the pre-registered criterion was satisfied in all three of its components, the effect survived correction for multiple comparisons, and showed no interaction with the choice of architecture (Experiment 1). It is a property of the configuration, which differs from the baseline in initialisation as well as in preprocessing.
4. The contribution of the individual pipeline stages was quantified through a cumulative component ablation under a single initialisation: every stage transition contributed more than the between-fold dispersion of its level, the contributions separate into groups with the two photometric stages leading, and the ablation recovered the whole in-domain gain. The CLAHE clip limit and the flat-field σ each exhibited an interior optimum within the ranges swept, confirmed on held-out data and supported by image-quality metrics (CNR, entropy, SSIM) (Experiment 2).
5. The reduction of source-to-target distance in feature space was measured directly across six external corpora under a mandatory source-domain-statistics condition: the distance fell on every corpus examined, supplying the middle term of the causal chain that the remaining investigations approach only through its consequence — in direction only, since the magnitude of a distance reduction does not track the magnitude of any performance gain.
6. A model trained on EyePACS with the pipeline transferred to APTOS 2019 without retraining, clearing the pre-registered generalisation ratio $G \geq 0.85$ and exceeding the baseline on every grade; the baseline also cleared the threshold, so the evidence lies in the comparison rather than in the criterion (Experiment 3).
7. Grad-CAM explainability analysis, using the proposed Attention–Lesion Overlap (ALO, primary) and IoU (secondary) against IDRiD pixel-level lesion masks, found model attention more closely aligned with expert annotation under the integrated configuration, on all four annotated lesion types and both measures (Experiment 4); the qualitative examination on the clinical corpus was not carried out.
8. On both external clinical corpora the integrated configuration attained higher absolute weighted F1, each difference reaching the minimal clinically important difference with its interval excluding zero (Experiment 5 — higher absolute external performance, not resistance to degradation); performance was maintained across all five camera groupings with their dispersion reduced (Experiment 6); and in the data-scarce regime the integrated configuration was

again the stronger, by a margin comparable to, and not larger than, the one measured with abundant data (Experiment 7).

9. A modular architecture for an automated DR screening system suitable for resource-limited environments, with telemedicine and eHealth integration applicable to Kazakhstan healthcare infrastructure, was designed, and a working demonstrator of it was built and deployed: it runs the preprocessing pipeline and the trained classifier end to end, produces attention maps, and keeps a per-case record including the reviewing ophthalmologist's verdict. The demonstrator establishes that the design is realisable and what it does in operation; the deployment-oriented parts it does not realise remain specification, and no clinical deployment testing was conducted.

Statements submitted for defense

Each statement is submitted at the strength its pre-specified criterion supports, and each carries a qualification that is not detachable from it: a statement given without its qualification is a different and stronger proposition than the evidence supports.

1. Preprocessing of fundus images is a formalisable and experimentally testable component of the diagnostic model rather than ancillary preparation of the data — the reframing from the end-to-end paradigm (P1) to the integrated paradigm (P2). *This statement is methodological and non-empirical: the experimental results are consistent with it under the conditions tested and do not establish it universally*.
2. The integrated configuration dominates the baseline configuration in five-class DR classification on EyePACS, on both ResNet-50 and EfficientNet-B3, under the conjunctive pre-registered criterion in all three of its components, with the effect surviving correction for multiple comparisons and showing no interaction with the choice of architecture. *The two arms differ in initialisation as well as in preprocessing, so the statement concerns the configuration as a whole; the cumulative ablation decomposes that composite but does not dissolve it, and no part of the effect is attributed to preprocessing alone.*
3. The dual-constraint CLAHE parameters and the flat-field σ each exhibit an interior optimum within the ranges examined, confirmed on held-out data. *Bounded to the ranges tested; the optimal values are properties of this corpus and pipeline rather than portable constants.*
4. The contributions of the individual pipeline stages are separable, monotone across folds, and orderable with the two photometric stages leading. *At the resolution of groupings only: adjacent ranks lie within noise, no strict ordering of the stages is submitted, and the FOV mask channel was not isolated.*
5. The integrated configuration reduces the distance between the training distribution and each external distribution in penultimate-layer feature space, on every external corpus examined, without any target-corpus statistic entering the transform. *In direction only — the magnitude of a distance reduction does not track the magnitude of any performance gain; the claim is mechanistic and carries no claim of diagnostic performance or device compatibility.*

6. A model trained on EyePACS with the pipeline transfers to APTOS 2019 without retraining, clearing the pre-registered generalisation ratio $G \geq 0.85$ and exceeding the baseline on every grade. *The baseline configuration also clears the threshold, so the criterion does not discriminate between the arms; the evidence lies in the comparison.*
7. Model attention aligns more closely with expert pixel-level lesion annotation under the integrated configuration, on all four annotated lesion types, on both the primary and the secondary measure, and robustly across the attention threshold. *Alignment between model evidence and expert annotation, not clinical localisation of pathology; one annotated corpus and one fold; the qualitative examination on the clinical corpus was not carried out.*
8. Classification performance is maintained across every camera grouping examined and, substantively, the dispersion of performance across those groupings is reduced. *Both configurations clear the retention floor; two of the five groupings are the external clinical corpora themselves and furnish no independent replication; three groupings aggregate several camera models rather than a single device; no part of this constitutes device certification or a claim of device-agnostic deployment readiness.*
9. The integrated configuration attains higher absolute weighted F1 than the baseline on both external clinical corpora, each difference reaching the minimal clinically important difference with its confidence interval excluding zero, evaluated on each corpus separately. *Higher absolute external performance, not resistance to degradation — relative to their own in-domain levels the two configurations decline almost identically; the margin over the minimal important difference on the second corpus is 0.0041, and the lower bound of that interval lies below the minimal important difference.*
10. Three measures in common use for expressing external robustness — the degradation quantity, the generalisation ratio and the retention ratio — share a structural defect: each normalises or differences an arm's external performance against that same arm's in-domain performance, and therefore penalises a configuration for its in-domain strength. *Analytic and strictly descriptive: it rehabilitates no result of this work.*
11. A coupled thermal-optical model of fundus tissue response under laser exposure, and a modular architecture for an automated DR screening system in resource-limited environments, realised in part as a working demonstrator. *Submitted as theoretical and design contributions respectively — the model is not clinically validated; the demonstrator establishes realisability and operating behaviour and is evidence for no diagnostic claim; the deployment-oriented parts of the architecture remain specification, and no field testing was conducted.*

One further result is offered as an observation rather than as a statement for defence: in a preregistered evaluation on a corpus roughly one-seventieth the size of the training corpus, with both arms trained from the same initialisation, the integrated configuration was again the stronger, by a margin **comparable to, and not larger**

than, the margin measured with abundant data. The clinical corpus involved cannot be redistributed, so that result is not externally reproducible.

The following are **not** submitted: clinical validity, readiness for unsupervised deployment, device certification, localisation of pathology, superiority over any named published system, and extension beyond the corpora, devices and hardware on which the results were obtained.

Theoretical significance

The work contributes a conceptual reframing of preprocessing as an integral model component that co-determines the feature space available to the CNN (paradigm P2), and places this reframing under direct empirical test against the end-to-end paradigm (P1). It provides a formal specification of the 8-stage pipeline and mathematical formalisations of the dual-constraint CLAHE clip limit — which extends the candidate's previously published work rather than originating here — of the adaptive flat-field correction, and of the ALO explainability metric, together with a statistical framework for validating preprocessing effects under class imbalance. A further contribution is of a different kind: a mechanism hitherto invoked as an explanation is made measurable, as a quantity computable at the penultimate layer under a stated protocol condition (normalisation from source-domain statistics), which makes the mechanistic claim falsifiable independently of the performance claims it explains. The coupled thermal-optical model of fundus tissue response under laser exposure is a theoretical and computational contribution only.

Practical significance

The integrated pipeline constitutes a fully specified, reproducible preprocessing regime for automated DR diagnosis under constrained computational conditions: it is defined stage by stage, with its operators, parameters and order of application, and its implementation is reproduced in Appendix A. The parameter values, however, were fixed on particular corpora and are properties of those corpora rather than portable constants.

A modular automated DR screening system architecture is proposed, with preprocessing-engine, inference, store-and-forward telemedicine, and physician-in-the-loop decision-support modules, integrable with PACS/EHR systems and the national eHealth platform, and applicable to DR screening in rural and underserved regions of Kazakhstan. A working demonstrator of the architecture is deployed and performs inference on submitted images, which establishes that the design is realisable and what it does in operation; the deployment-oriented parts it does not realise — PACS integration, EHR interoperability, telemedicine operation — remain design specifications, no deployment was tested clinically, and the data-protection provisions are alignment by design rather than certified compliance. The system is conceived as decision support in which the clinician retains the diagnosis and the responsibility for it; nothing here is offered as a standalone diagnostic instrument.

Reliability of the results

Reliability is ensured by: the use of large-scale and multiple public datasets (~35,126 EyePACS training images and seven additional datasets); 5-fold patient-level stratified cross-validation with strict leakage control; reporting of all primary metrics as mean ± standard deviation; a multi-metric assessment (weighted F1, ROC-AUC, Cohen's Kappa with quadratic weights, accuracy); statistical hypothesis testing (McNemar, DeLong), mixed-effects modelling across folds, bootstrap confidence intervals (≥ 1000 resamples), and Bonferroni/Holm correction; and success criteria fixed before the experiments that tested them — a criterion written once a result is known can always be satisfied, so the ordering is what makes the classification informative. The work is additionally placed against published systems without being ranked among them, since no two of the compared results share their task definition, corpus, reference standard and metric. Two limits of this apparatus are stated rather than reserved: correction for multiple comparisons is scoped to the single experiment within which the comparisons were planned, so no dissertation-wide error rate is claimed; and several evaluations rest on the models of one fitted fold, so their intervals quantify sampling of the evaluation corpus alone and understate total uncertainty in a known direction. A third is stated with them: one experiment depends on a clinical corpus that cannot be redistributed, so that result is not externally reproducible, and calibration — measured and reported — is an empirical property of a model's output probabilities, not a warrant of clinical decision-making reliability.

Approbation of the results and connection with scientific programmes

The results of the research were reported and discussed at international scientific forums, including the 3rd International Workshop on Digital Society (DS 2025, Istanbul, Türkiye, 28–30 October 2025), and published in international peer-reviewed journals; the five works are listed below.

The research direction corresponds to the state priorities of healthcare digitalisation and the development of artificial-intelligence technologies of the Republic of Kazakhstan, in particular: the Concept for the Development of Artificial Intelligence for 2024–2029; the Address of the President of the Republic of Kazakhstan «Kazakhstan in the Era of Artificial Intelligence» (8 September 2025); and the Law of the Republic of Kazakhstan «On Artificial Intelligence» (No. 230-VIII, 17 November 2025). The work is carried out in accordance with subpara. 2, para. 3, art. 20 of the Law of the Republic of Kazakhstan «On Science».

This is a correspondence between the direction of the research and published national priorities: it is not a statement that the work was funded under a state programme, commissioned by any body, or performed under an institutional mandate, nor a claim that any policy objective has been achieved by it.

Publications

The main results of the dissertation are published in 5 scientific works, including: articles in journals indexed by Scopus / Web of Science — 1 (*Eastern-European Journal of Enterprise Technologies*, Q3); a paper in international conference proceedings indexed by Scopus — 1 (*Procedia Computer Science*, DS 2025); and articles in journals recommended by the Committee for Quality Assurance in Science and Higher Education (KKSON) of the Republic of Kazakhstan — 3 (*News of the NAS RK, Physico-Mathematical Series*; *Herald of the Kazakh-British Technical University*; *Herald of KazUTB*).

Articles in journals indexed by Scopus / Web of Science:

1. Sapakova S., Yesmukhamedov N., Sapakov A. Development of an image quality enhancement approach for diabetic retinopathy diagnosis // Eastern-European Journal of Enterprise Technologies. — 2025. — Vol. 4, No. 9(136). — P. 79–88. (Scopus, Q3). https://doi.org/10.15587/1729-4061.2025.335570

Papers in international conference proceedings indexed by Scopus:

2. Sapakova S., Yesmukhamedov N., Sapakov A., Yemberdiyeva A., Kozhamkulova Z. Methods for pre-processing and analysis of fundus images for detection of diabetic retinopathy // Procedia Computer Science. — 2025. — Vol. 272. — P. 496–501. (The 3rd International Workshop on Digital Society — DS 2025, Istanbul, Türkiye). https://doi.org/10.1016/j.procs.2025.10.237

Articles in journals recommended by the KKSON of the Republic of Kazakhstan:

3. Yesmukhamedov N.S., Sapakova S., Al-Haddad S.A.R., Daniyarova D. Development of an information system architecture for healthcare institutions using artificial intelligence // News of the National Academy of Sciences of the Republic of Kazakhstan, Physico-Mathematical Series. — 2025. — Vol. 2(354). — P. 74–91. https://doi.org/10.32014/2025.2518-1726.345
4. Yesmukhamedov N.S., Sapakova S.Z., Kozhamkulova Zh.Zh., Daniyarova D.R., Armankyzy R. Methods for preprocessing and analysis of fundus images for diabetic retinopathy detection // Herald of the Kazakh-British Technical University. — 2025. — No. 4(75). — Vol. 22. — P. 119–130. https://doi.org/10.55452/1998-6688-2025-22-4-119-130
5. Sapakova S.Z., Daniyarova D.R., Yesmukhamedov N.S., Armankyzy R., Yemberdiyeva A.B., Kaldybaeva A.S. Mathematical modeling of laser exposure on fundus tissues in the treatment of diabetic retinopathy // Herald of KazUTB. — 2025. — Vol. 2, No. 27-740. https://doi.org/10.58805/kazutb.v.2.27-740

Main content of the work

The introduction substantiates the relevance of the topic, formulates the aim, objectives, object and subject of the research, the central hypothesis, the scientific novelty, the statements submitted for defense, the methodological basis, the theoretical and practical significance, the reliability of the results, the empirical basis, the approbation, the connection with scientific programmes, and the publications, and describes the structure and length of the dissertation.

Chapter 1 — "Automated diabetic retinopathy screening" establishes the clinical and technical context of screening: the ordinal five-class grading of the disease and the clinical cost of a misgrading, the screening requirements of resource-limited settings, the physics of laser coagulation as bounded theoretical context for referral, the sources of fundus image quality loss and its device-specific component, what the field has achieved with convolutional networks (architectures, transfer and self-supervised pretraining, explainability), and a critical analysis of existing automated screening systems and the conditions that bound their reported figures, concluding with the formulation of the research problem and the justification of the research direction.

Chapter 2 — "Methodology of the integrated pipeline" specifies the integrated methodology. It formalises the 8-stage preprocessing pipeline stage by stage together with the theoretical foundations of each — canonical orientation and landmark-based rotation normalisation, isotropic resize with centred padding, the field-of-view mask as a fourth input channel, adaptive flat-field correction scaled to per-image geometry, and the dual-constraint clip limit of the CLAHE stage, formalised as the minimum of a histogram-relative and a tile-relative constraint and applied stochastically at training time. It then specifies the classification architectures (ResNet-50 and EfficientNet-B3) and their adaptation to five-class grading with the weighted (Focal) loss, the pretraining and fine-tuning strategy with its frozen-backbone linear-probe acceptance gate, the explainability formalism (Grad-CAM, the Attention–Lesion Overlap metric, IoU) with the image-quality measures, and the evaluation hierarchy and statistical protocol.

Chapter 3 — "Experimental results" reports the experimental programme on the tiered corpus: the datasets and experimental configuration; the effect of the pipeline on accuracy under the 2×2 factorial design (Experiment 1); the cumulative stage ablation with the CLAHE clip-limit and flat-field σ sweeps (Experiment 2); the direct measurement of source-to-target distance in penultimate-layer feature space across six external corpora; cross-dataset transfer to APTOS 2019 and external clinical evaluation on IDRiD and Messidor-2 (Experiments 3 and 5), with behaviour across camera groupings (Experiment 6); the agreement of attention maps with expert pixel-level lesion annotation (Experiment 4); training on small clinical samples (Experiment 7); and the statistical validation and the comparative placement of the results against published systems without ranking among them, closing with the limitations and boundary conditions of the approach.

Chapter 4 — "The screening system" describes the screening system built around the model, distinguishing throughout what was realised from what remains specification: the modular architecture and its modules, the preprocessing and inference services with their interface, the clinical workflow and the operator interface in which the ophthalmologist reviews and records a verdict, and the deployment envelope and data-protection framework (GDPR/HIPAA-aligned by design), with the applicability to Kazakhstan healthcare infrastructure bounded by the absence of field testing there.

Author's personal contribution

The principal results of the dissertation were obtained by the author personally: the conceptual reframing of preprocessing as an integral component of the diagnostic model (paradigm P2); the design and formal specification of the 8-stage preprocessing pipeline, including the dual-constraint stochastic CLAHE, the adaptive flat-field correction, and the FOV-mask input channel; the Attention–Lesion Overlap (ALO) explainability metric; the controlled experimental design and statistical validation framework; and the architecture of the automated DR screening system. In the joint publications the author developed the fundus image preprocessing and analysis methods and prepared the corresponding parts of the manuscripts.

Structure and length of the dissertation

The dissertation consists of front matter — normative references, definitions, and designations and abbreviations — followed by an introduction, four chapters, a conclusion, a list of references used, and five appendices (A–E): the source code of the preprocessing pipeline, supplementary experimental results and confusion matrices, the system-architecture diagrams, the attention-map gallery, and supplementary tables for the device-domain evaluation.

The dissertation is set out on 117 pages, and contains 19 tables and 16 figures. The list of references comprises 99 sources. The counts are of the dissertation excluding the appendices.